Math Hard Question 13

1 ★ Mark for Review ABC

A circle has center O, and points A and B lie on the circle. The measure of arc AB is $60°$, and the length of this arc is 4 inches. What is the circumference, in inches, of the circle?

A. 4

B. 8

C. 16

D. 24

2 ★ Mark for Review ABC

In the xy-plane, the graph of function f, where $y = f(x)$, has exactly 8 x-intercepts. One of these x-intercepts is $(17, 0)$. The rational function g is defined by:

$$g(x) = \frac{f(x)}{x - 17}$$

In the xy-plane, how many x-intercepts does the graph of $y = g(x)$ have?

A. 5

B. 6

C. 7

D. 8

3 ★ Mark for Review ABC

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week:

- The number of prey had increased by 2700% of the number of prey at the start of the week.
- The number of predators had increased by 180% of the number of predators at the start of the week.

The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week. What is the value of p?

A. 9

B. 90

C. 900

D. 9000

4 ★ Mark for Review ABC

The exponential function f is defined by

$$f(x) = ab^x,$$

where a and b are positive constants. If $f(s) = t$ and $f(s+1) = t - 0.87t$, where s and t are constants, what is the value of b?

A. 0.13

B. 0.87

C. 1.13

D. 1.87

5 ★ Mark for Review **ABC**

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees.

The beekeeper uses a model that predicts the population of this bee colony:

- begins at 1,100
- increases by 120 bees per week in the first two weeks
- then increases by 180 bees per week until the goal is reached

According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

A. $p(w) = 2600 - 180w$

B. $p(w) = 2480 + 180w$

C. $p(w) = 1620 - 180w$

D. $p(w) = -120 + 180w$

6 ★ Mark for Review ABC

Given the quadratic equation:

$$7rx^2 - 28sx + 7r$$

In the equation:

- r is a positive constant
- s is a negative constant
- The equation has two solutions

If $\frac{r}{s} > k$, what is the minimum value of k?

7 ★ Mark for Review ABC

A square map has a side length of 55 inches, and 1 inch on the map represents an actual distance of 13 miles. A larger version of the same map is printed as a square with the side length 90% longer than the side length of the previous map. On the larger map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A. 1.30

B. 6.84

C. 24.70

D. 49.50

8 ★ Mark for Review **ABC**

In isosceles triangle ABC, sides AB and AC are congruent. Point D divides side $\overline{BC}$ such that the length of $\overline{BD}$ is $\dfrac{4}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments $\overline{DE}$ and $\overline{DF}$ are drawn, $\angle BED$ is congruent to $\angle CFD$. If the length of $\overline{BE}$ is 8, what is the length of $\overline{CF}$?

A. 6

B. 14

C. 32

D. 56

9 ★ Mark for Review **ABC**

$$\frac{\sqrt[3]{x^{17}y^5}}{5x^2\left(\sqrt[8]{y^8}\right)}$$

For all positive values of x and y, the given expression is equivalent to which of the following?

1. $\dfrac{x^{\frac{14}{3}}y^{\frac{5}{3}}}{5xy}$
2. $\dfrac{\sqrt[3]{x^{11}y^2}}{5}$

A. I only

B. II only

C. I and II

D. Neither I nor II

10 ★ Mark for Review ABC

The table shows three values of x and their corresponding values of y, where s is a constant. There is a linear relationship between x and y. Which of the following equations represents this relationship?

x	y
$-2s$	24
$-s$	21
s	15

A. $sx + 3y = 18s$

B. $3x + sy = 18s$

C. $3x + sy = 18$

D. $sx + 3y = 18$

11 ★ Mark for Review ABC

The cost of renting a carpet cleaner is \$52 for the first day and \$26 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

A. $C(d) = 26d + 26$

B. $C(d) = 26d + 52$

C. $C(d) = 52d - 26$

D. $C(d) = 52d + 78$

12 ★ Mark for Review ABC

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in **miles per minute squared**, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

A. 0.3

B. 16.3

C. 195.8

D. 220.4

13 ★ Mark for Review ABC

The table shows three values of x and their corresponding values of $g(x)$, where

$$g(x) = \frac{f(x)}{x+5}$$

and f is a linear function. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

x	$g(x)$
-20	4
-8	0
10	6

A. $(0, -8)$

B. $(0, 5)$

C. $(0, 8)$

D. $(0, 40)$